AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2018

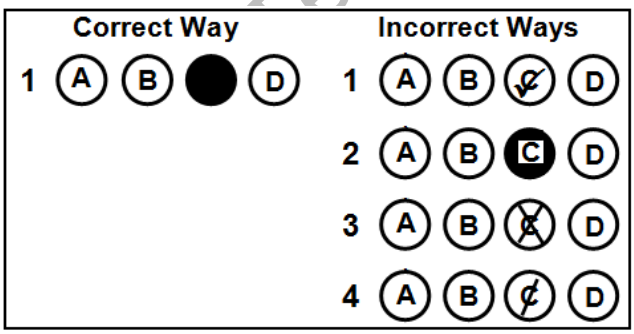
General Mathematics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. 8% of 200 is equal to

   A. 4
   B. 8
   C. 16
   D. 25

2. A chocolate box contains 250 chocolates. If some children eat 70% of these chocolates, then the remaining chocolates will be

   A. 75
   B. 150
   C. 175
   D. 220

3. The ratio of the ages of Nazia and Shazia is 2 : 3. If Nazia is 10 years old, then Shazia's age is

   A. 12 years.
   B. 13 years.
   C. 15 years.
   D. 18 years.

4. A man leaves behind a land worth Rs 300,000. If he has two sons and a daughter as his legal heirs, then the inheritance of daughter is

   (Note: The share of son is two times that of daughter.)

   A. Rs 60,000
   B. Rs 80,000
   C. Rs 100,000
   D. Rs 120,000

5. Majid earns Rs 225,000 from his rice field and Rs 300,000 from his cotton field in a certain year. The amount of Ushr he has to pay this year is

   (**Note:** Rate of Ushr is 10% of the earning.)

   A. Rs 5,250
   B. Rs 22,500
   C. Rs 30,000
   D. Rs 52,500

6. In a sale, a shopkeeper reduced the marked price of the goods by 15%. If the marked price of a diary was Rs 500, then its price in sale is

   A. Rs 425
   B. Rs 485
   C. Rs 515
   D. Rs 575

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7. A gold chain bought for Rs 50,000 is sold after two years at a profit of Rs 9,500. The cost of this chain has increased by

   A. 19%
   B. 23.5%
   C. 119%
   D. 123.5%

8. *A* and *B* are business partners. They made a profit of Rs 60,000. If *A* has double the share as compared to *B*, then the share of profit for *A* is equal to

   (**Note:** Profit sharing ratio is on the basis of amount invested by each partner.)

   A. Rs 20,000
   B. Rs 30,000
   C. Rs 40,000
   D. Rs 50,000

9. A person insured his car at an assessed market price of Rs 830,000 at an annual premium of 7%. The amount of annual premium will be

   A. Rs 5,810
   B. Rs 6,889
   C. Rs 58,100
   D. Rs 68,890

10. Sara deposited Rs 22,000 in her account. If the bank pays simple interest at the rate of 4% per year, the interest she gets after 5 years will be

    A. Rs 4,000
    B. Rs 4,400
    C. Rs 40,000
    D. Rs 44,000

11. A car rental service charges Rs 550 for the first 6 kilometres and Rs 20 for each additional kilometre. If Abid travels 25 km using this car service, then he has to pay

    A. Rs 570
    B. Rs 930
    C. Rs 1,050
    D. Rs 1,170

12. A laptop scheme allows students to purchase a laptop on 50% down payment and the remaining amount has to be paid in 2 equal installments. If a laptop costs Rs 60,000, then each installment will be

    A. Rs 15,000
    B. Rs 20,000
    C. Rs 30,000
    D. Rs 45,000

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13. Wahid pays income tax of Rs 6,000 in February and his taxable income for February is Rs 120,000. The levied rate of income tax is

A. 0.2%
B. 0.5%
C. 2%
D. 5%

14. A cellular company pays Rs 200 per hour and an additional commission of Rs 30 on the sale of each SIM to its sales person. If a sales person sold 6 SIMs in 5 hours, then the total amount the company will pay him is

A. Rs 1,030
B. Rs 1,180
C. Rs 1,230
D. Rs 1,350

15. A receptionist works 8 hours a day. If she earns Rs 185 per hour, then her salary for a week is

(**Note:** The week consists of 5 working days.)

A. Rs 925
B. Rs 1,057
C. Rs 1,295
D. Rs 7,400

16. When the annual income exceeds Rs 400,000, income tax of 5% is applicable on the exceeding amount. The annual income tax of a person having an annual salary of Rs 600,000 is

A. Rs 10,000
B. Rs 20,000
C. Rs 30,000
D. Rs 50,000

17. The exponent in the equation $(x+y)^w = z$ is

A. 1
B. z
C. w
D. $x+y$

18. The ordinary form (common form) of the scientific notation 5.38×10^{-2} is

A. 0.00538
B. 0.0538
C. 538
D. 5380

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19. The exponential form of $\log_3 5 = 2x$ is

A. $(2x)^3 = 5$
B. $(2x)^5 = 3$
C. $5^{2x} = 3$
D. $3^{2x} = 5$

20. The 5th term of the geometric progression 3, 9, 27, 81 is

A. 90
B. 99
C. 243
D. 729

21. The number of terms in the geometric progression 1, 3, 3^2, 3^3, ..., 3^6 is

A. 5
B. 6
C. 7
D. 8

22. The general term of the sequence 10, 5, 0, − 5, − 10 is

A. $T_n = 10(2)^{n-1}$
B. $T_n = 10\left(\frac{1}{2}\right)^{n-1}$
C. $T_n = 10 + 5(n-1)$
D. $T_n = 10 - 5(n-1)$

23. If a universal set is defined as U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} and E is set of even number, then E' is

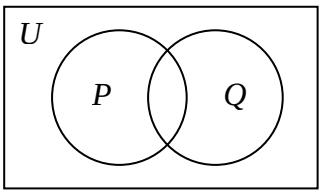
A. ϕ
B. {1, 3, 5, 7, 9}
C. {2, 4, 6, 8, 10}
D. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}

24. If X = {10, 20} and Y = {p, q, r}, then which of the following represents a function from X to Y?

A. {(10, p)}
B. {(10, p), (10, q)}
C. {(10, p), (20, p)}
D. {(10, p), (10, q), (20, r)}

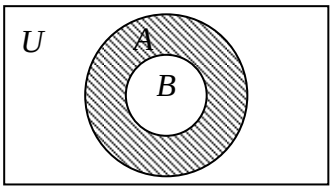
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25. For the given Venn diagram, $P \cup (Q \cap P)$ is equal to



A. P
B. Q
C. $P - Q$
D. $Q - P$

26. The shaded area in the given Venn diagram represents



A. $A \cup B$
B. $A \cap B$
C. $B - A$
D. $A - B$

27. The point $(-8, 5)$ lies in the

A. I quadrant.
B. II quadrant.
C. III quadrant.
D. IV quadrant.

28. Which of the following points lie on the line $y = 2x$?

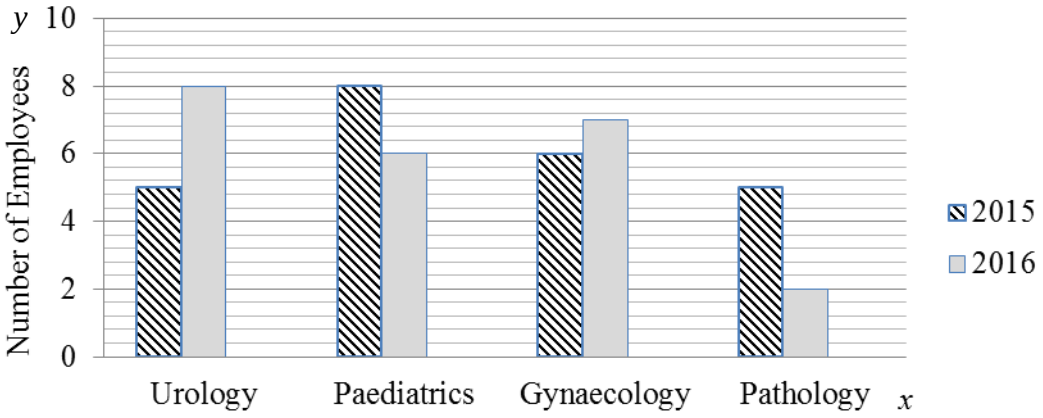
A. (1, 1)
B. (1, 2)
C. (2, 1)
D. (2, 2)

29. The ages (in years) of nine children in a group are 4, 7, 8, 9, 6, 10, 5, 8 and 10. The median age is

A. 6
B. 7
C. 8
D. 10

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30. The given bar chart shows the number of employees hired by a hospital in various departments in the years 2015 and 2016.



In Paediatrics department, the percentage change shows that hiring in 2016 as compared to last year has

A. decreased by 25%
B. decreased by 75%
C. increased by 25%
D. increased by 75%

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